

Project Report: SpendSmart Expense Tracker

Course: ICS 2300 Mobile Applications Design and Development

Group: Group 2

Technical Lead: Shemiramoth Mugo

1. Project Overview & Justification

The **SpendSmart** application is a modern personal finance tool designed to solve the real-world problem of "invisible spending" among young adults. By providing a categorized logging system and visual progress bars, the app encourages financial literacy and prevents over-expenditure. The official project repository is maintained at: <https://github.com/Shemira01/MADD-Project-2.git>.

2. Global App Architecture

The application is built using **Java** and follows a robust development structure:

- **Structure:** Single Activity Architecture where MainActivity serves as the primary host.
 - **Navigation:** A BottomNavigationView using a dedicated Menu Resource for seamless transitions.
 - **Fragment Management:** Dynamic swapping using getSupportFragmentManager and replace() transactions.
 - **Design Language:** Material Design 3 (M3) utilizing TextInputLayout.OutlinedBox and custom color mapping (colorPrimary to primary) for a modern, professional interface.
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3. Page-by-Page Breakdown

Page 1: Dashboard (DashboardFragment)

- **UI Components:** MaterialCardView for Balance display and a ProgressBar for visual budget status.
- **Content:** A RecyclerView to display recent transactions.

Page 2: Add Expense (AddExpenseFragment)

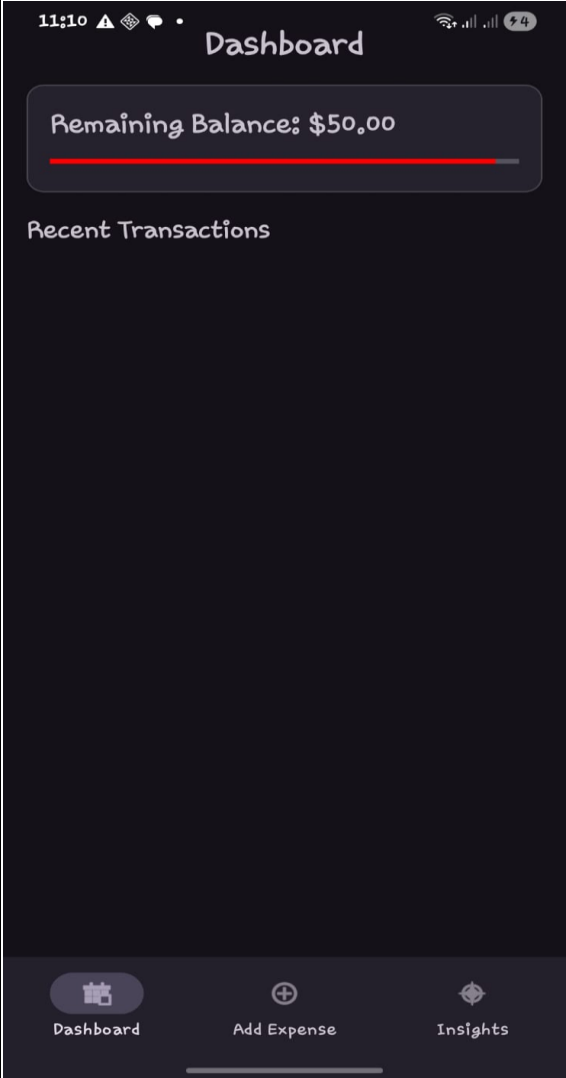
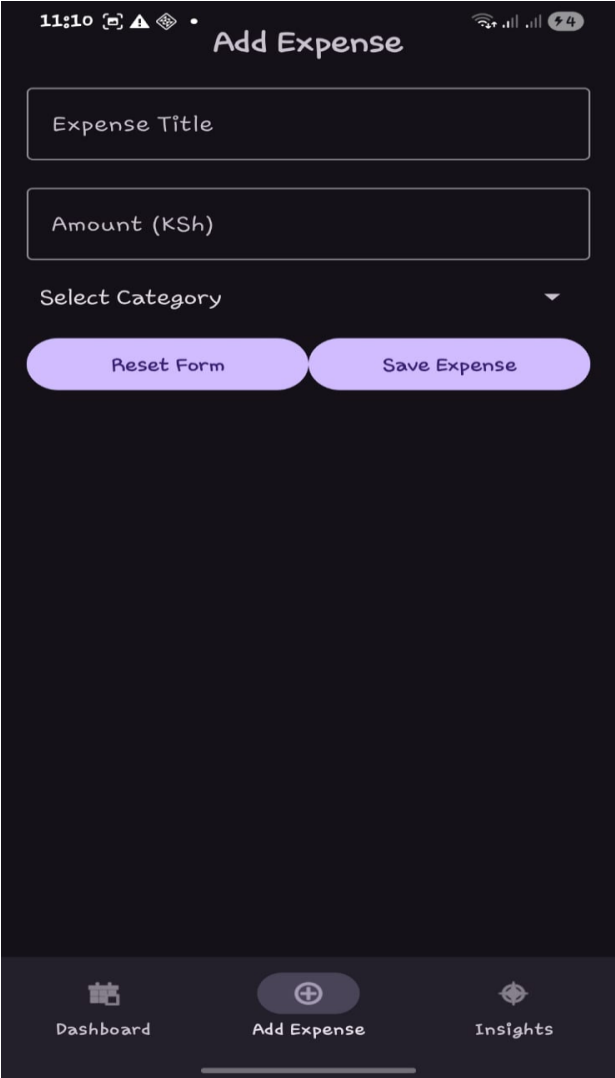
- **UI Components:** TextInputLayout (Outlined style), TextInputEditText, and a Spinner for categories.
- **Key Feature:** resetForm() – This programmatically clears all inputs and returns cursor focus to the amount field to facilitate high-speed data entry.

Page 3: Insights (InsightsFragment)

- **UI Components:** A statistical breakdown of spending categories using horizontal progress indicators to show budget consumption.

4. Implementation & Physical Device Testing

The application was successfully deployed and tested on a physical mobile device to verify Material 3 responsiveness and form logic.

Figure 1: Dashboard Interface  The screenshot shows the 'Dashboard' screen of a mobile application. At the top, the status bar displays the time 11:10 and various icons. Below the status bar, the word 'Dashboard' is centered. A dark grey rounded rectangle contains the text 'Remaining Balance: \$50.00' above a red horizontal progress bar. Below this, the text 'Recent Transactions' is visible. At the bottom, there is a navigation bar with three icons: a house for 'Dashboard', a plus sign for 'Add Expense', and a magnifying glass for 'Insights'.	Figure 2: Add Expense Interface  The screenshot shows the 'Add Expense' screen. At the top, the status bar displays the time 11:10 and various icons. Below the status bar, the text 'Add Expense' is centered. There are three input fields: 'Expense Title', 'Amount (KSh)', and 'Select Category' with a dropdown arrow. Below the input fields are two purple buttons: 'Reset Form' and 'Save Expense'. At the bottom, there is a navigation bar with three icons: a house for 'Dashboard', a plus sign for 'Add Expense', and a magnifying glass for 'Insights'.
<i>The Dashboard showing the \$50.00 balance and budget consumption progress bar.</i>	<i>The 'Add Expense' form with Kenyan Shilling (KSh) formatting and category selection.</i>

5. Team Roles & Task Allocation

Member Name	Project Role	Technical Deliverable
Shemiramoth Mugo	Technical Lead	Architecture Backbone, Fragment Skeleton, & UI Reset Logic.
Lee Kariuki	Lead UI Designer	Material Design 3 Styling (Themes/Colors) & XML Layouts.
Gabriella Muthoni	Input System Dev	Entry Form Data capture & Object Mapping.
Felix Wanyoike	Validation Lead	Field Error Handling & Currency Formatting Logic.
Cecil Kioko	Database Manager	Local Data Persistence (Room/SQLite) - CRUD Operations.
Emmanuel Mwangangi	Math & Logic Dev	Spending Analysis, Budget Math, & Category Percentages.
Linnet Mungai	Visualization Dev	Dashboard Status UI & Insights Visual Breakdown.
Samuel Kuria	QA & Tech Writer	Unit Testing & Final Technical Report.

6. Conclusion

The **SpendSmart** project successfully transitioned from a conceptual blueprint to a functional mobile application. By utilizing Java and the Single Activity Architecture, the team delivered a tool that effectively visualizes personal finances. Deployment on a physical device confirmed that the UI remains intuitive and the data validation logic is stable for real-world use.

Prepared by: Samuel Kuria.